Overview: The Radiative Transfer Equation and Chandrasekhar's Formal Solution

Notes for EE 445/645 — Physical Models in Remote Sensing

1 The Radiative Transfer Equation

The RTE is fundamentally a **photon balance equation**. Along a ray in direction $\mathbf{\Omega}$, the change in radiance I per unit path length s equals gains minus losses:

$$\frac{dI}{ds} = \underbrace{-\kappa_e\, I}_{\text{extinction}} + \underbrace{\kappa_s \int_{4\pi} p(\mathbf{\Omega}' \to \mathbf{\Omega})\, I(\mathbf{\Omega}')\, d\Omega'}_{\text{inscattering}} + \underbrace{\kappa_a\, B(T)}_{\text{emission}} \tag{1}$$

where:

- $\kappa_e = \kappa_a + \kappa_s$ extinction coefficient (absorption + scattering)
- $p(\mathbf{\Omega}' \to \mathbf{\Omega})$ phase function, normalized so that $\frac{1}{4\pi}\int_{4\pi} p\, d\Omega = 1$
- $B(\lambda, T)$ Planck blackbody spectral radiance
- $\omega = \kappa_s/\kappa_e$ single scattering albedo ($\omega = 0$: pure absorption; $\omega = 1$: pure scattering)

Plane-Parallel Form (Myneni's Geometry)

For a horizontally uniform medium with z increasing downward and directional cosine $\mu = \cos\theta$, the steady-state RTE becomes:

$$-\mu\, \frac{\partial I_\lambda(z, \mathbf{\Omega})}{\partial z} + \sigma(z, \mathbf{\Omega})\, I_\lambda(z, \mathbf{\Omega}) = \int_{4\pi} \sigma_{s\lambda}(z, \mathbf{\Omega}' \to \mathbf{\Omega})\, I_\lambda(z, \mathbf{\Omega}')\, d\Omega' \tag{2}$$

with boundary conditions:

$$I_\lambda(z = 0, \mathbf{\Omega}) = I_{0\lambda}\, \delta(\mathbf{\Omega} - \mathbf{\Omega}_0) + I_{d\lambda}(\mathbf{\Omega}), \quad \mu < 0 \quad \text{(top: sun + diffuse sky)} \tag{3}$$

$$I_\lambda(z = z_H, \mathbf{\Omega}) = 0, \quad \mu > 0 \quad \text{(bottom: no upwelling source below canopy)} \tag{4}$$

2 Chandrasekhar's Formal Solution

2.1 Setup

Consider the simplified case — a single ray with absorption and a general source $j(s)$ (which, when scattering is present, will itself contain an integral over the radiation field):

$$\frac{dI}{ds} + \kappa(s)\, I = j(s) \tag{5}$$

This is a first-order linear ODE. Define the **optical depth** between two points:

$$\tau(s_1, s_2) = \int_{s_1}^{s_2} \kappa(s')\, ds' \tag{6}$$

τ is dimensionless and measures how "opaque" the medium is between s_1 and s_2. A value $\tau \gg 1$ means the medium is optically thick (most radiation absorbed or scattered); $\tau \ll 1$ means optically thin (radiation passes through relatively unimpeded).

2.2 Derivation via Integrating Factor

The integrating factor is:

$$\mu(s) = e^{\tau(0,\, s)} = \exp\left(\int_0^s \kappa(s')\, ds'\right) \tag{7}$$

Multiplying both sides of the ODE:

$$\frac{d}{ds}[\mu(s)\, I(s)] = \mu(s)\, j(s) \tag{8}$$

Integrating from 0 to s:

$$\mu(s)\, I(s) - I(0) = \int_0^s \mu(s')\, j(s')\, ds' \tag{9}$$

Dividing through by $\mu(s) = e^{\tau(0,s)}$ yields the **formal solution**:

$$\boxed{I(s) = I(0)\, e^{-\tau(0,\, s)} + \int_0^s j(s')\, e^{-\tau(s',\, s)}\, ds'} \tag{10}$$

2.3 Physical Interpretation

Each term has a clear physical meaning:

Term	**Physical Meaning**
$I(0)\, e^{-\tau(0,s)}$	Boundary radiance at $s = 0$, attenuated by Beer's Law as it propagates to s. The factor $e^{-\tau}$ is the *transmittance*.
$\int_0^s j(s')\, e^{-\tau(s',s)}\, ds'$	Every source element $j(s')\, ds'$ along the path contributes to $I(s)$, weighted by how much it is attenuated ($e^{-\tau(s',s)}$) before reaching s.

2.4 The Scattering Complication

When scattering is included, the source term becomes:

$$j(s) = \kappa_s(s) \int_{4\pi} p(\mathbf{\Omega}' \to \mathbf{\Omega})\, I(s, \mathbf{\Omega}')\, d\Omega' + \kappa_a(s)\, B(T(s)) \tag{11}$$

This makes the RTE **integro-differential**: the source at every point depends on the radiation field everywhere else. The formal solution is still valid, but j itself depends on I, so the equation cannot be solved in closed form in general.

This is precisely why Chapter 4 of the course is devoted entirely to numerical methods:

- **Successive orders of scattering** — iterate, adding one scattering order at a time
- **Discrete ordinates (DOM)** — discretize directions $\mathbf{\Omega}$ into a finite set; Chandrasekhar's own method
- **Gauss-Seidel iteration** — solve the discretized system iteratively
- **Two-stream approximation** — reduce to upward/downward fluxes only; analytically tractable

3 Summary

The RTE is an energy balance: changes in radiance along a ray equal inscattering and emission gains minus extinction losses. Chandrasekhar's formal solution expresses $I(s)$ as a superposition of the attenuated boundary condition and all source contributions along the path, each weighted by the transmittance $e^{-\tau}$. When scattering couples all directions together, numerical methods are required to close the system.